Comparativa de Semánticas Formales (NS vs SOS)

1. Semántica Natural (NS)

$$\mathcal{T}_1 = \langle x \ := \ 1, \ \sigma_0\rangle \longrightarrow \sigma_0[x \mapsto \mathcal{A}[\![1]\!]\sigma_0] \ [\text{ass}_{\text{ns}}]$$

$$\mathcal{T}_2 = \langle y \ := \ 5, \ \sigma_1\rangle \longrightarrow \sigma_1[y \mapsto \mathcal{A}[\![5]\!]\sigma_1] \ [\text{ass}_{\text{ns}}]$$

$$\mathcal{T}_3 = \frac{\langle \varepsilon, \ \sigma_3\rangle \longrightarrow_D \sigma_3 \ [\text{none}_{\text{ns}}]}{\langle \textbf{var} \ y \ := \ 2\,;, \ \sigma_2\rangle \longrightarrow_D \sigma_2[y \mapsto \mathcal{A}[\![2]\!]\sigma_2]} \ [\text{var}_{\text{ns}}]$$

$$\mathcal{T}_4 = \langle x \ := \ x \ + \ 5, \ \sigma_3\rangle \longrightarrow \sigma_3[x \mapsto \mathcal{A}[\![x \ + \ 5]\!]\sigma_3] \ [\text{ass}_{\text{ns}}]$$

$$\mathcal{T}_5 = \langle x \ := \ 0, \ \sigma_4\rangle \longrightarrow \sigma_4[x \mapsto \mathcal{A}[\![0]\!]\sigma_4] \ [\text{ass}_{\text{ns}}]$$

$$\mathcal{T}_6 = \langle y \ := \ y \ + \ 1, \ \sigma_5\rangle \longrightarrow \sigma_5[y \mapsto \mathcal{A}[\![y \ + \ 1]\!]\sigma_5] \ [\text{ass}_{\text{ns}}]$$

$$\mathcal{T}_7 = \langle x \ := \ x \ + \ 1, \ \sigma_6\rangle \longrightarrow \sigma_6[x \mapsto \mathcal{A}[\![x \ + \ 1]\!]\sigma_6] \ [\text{ass}_{\text{ns}}]$$

$$\mathcal{T}_8 = \langle y \ := \ y \ + \ 1, \ \sigma_7\rangle \longrightarrow \sigma_7[y \mapsto \mathcal{A}[\![y \ + \ 1]\!]\sigma_7] \ [\text{ass}_{\text{ns}}]$$

$$\mathcal{T}_9 = \langle \textbf{for} \ x \ := \ x \ + \ 1 \ \textbf{to} \ 1 \ \textbf{do} \ y \ := \ y \ + \ 1, \ \sigma_8\rangle \longrightarrow \sigma_8 \ [\text{for} - \text{ff}_{\text{ns}}]$$

$$\mathcal{T}_{10} = \frac{\mathcal{T}_7 \qquad \mathcal{T}_8 \qquad \mathcal{T}_9}{\langle \textbf{for} \ x \ := \ x \ + \ 1 \ \textbf{to} \ 1 \ \textbf{do} \ y \ := \ y \ + \ 1, \ \sigma_6\rangle \longrightarrow \sigma_8} \ [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{11} = \frac{\mathcal{T}_5 \qquad \mathcal{T}_6 \qquad \mathcal{T}_{10}}{\langle \textbf{for} \ x \ := \ 0 \ \textbf{to} \ 1 \ \textbf{do} \ y \ := \ y \ + \ 1, \ \sigma_4\rangle \longrightarrow \sigma_8} \ [\text{for} - \text{tt}_{\text{ns}}]$$

$$\mathcal{T}_{12} = \frac{\mathcal{T}_4 \qquad \mathcal{T}_{11}}{\langle S_1, \ \sigma_3\rangle \longrightarrow \sigma_8} \ [\text{comp}_{\text{ns}}]$$

$$\mathcal{T}_{13} = \frac{\mathcal{T}_3 \qquad \mathcal{T}_{12}}{\langle S_2, \ \sigma_2\rangle \longrightarrow \sigma_8[\text{DV}(\textbf{var} \ y \ := \ 2\,;) \longmapsto \sigma_2]} \ [\text{block}_{\text{ns}}]$$

$$\mathcal{T}_{14} = \frac{\mathcal{T}_2 \qquad \mathcal{T}_{13}}{\langle S_3, \ \sigma_1\rangle \longrightarrow \sigma_2} \ [\text{comp}_{\text{ns}}]$$

$$\mathcal{T}_{15} = \frac{\mathcal{T}_1 \qquad \mathcal{T}_{14}}{\langle S_4, \ \sigma_0\rangle \longrightarrow \sigma_2} \ [\text{comp}_{\text{ns}}]$$

Sentencias Abreviadas

$$S_1 = (x \ := \ x \ + \ 5\,; \ \textbf{for} \ x \ := \ 0 \ \textbf{to} \ 1 \ \textbf{do} \ y \ := \ y \ + \ 1)$$
$$S_2 = \textbf{begin var} \ y \ := \ 2\,; \ (x \ := \ x \ + \ 5\,; \ \textbf{for} \ x \ := \ 0 \ \textbf{to} \ 1 \ \textbf{do} \ y \ := \ y \ + \ 1) \ \textbf{end}$$
$$S_3 = (y \ := \ 5\,; \ \textbf{begin var} \ y \ := \ 2\,; \ (x \ := \ x \ + \ 5\,; \ \textbf{for} \ x \ := \ 0 \ \textbf{to} \ 1 \ \textbf{do} \ y \ := \ y \ + \ 1) \ \textbf{end})$$
$$S_4 = (x \ := \ 1\,; \ y \ := \ 5\,; \ \textbf{begin var} \ y \ := \ 2\,; \ (x \ := \ x \ + \ 5\,; \ \textbf{for} \ x \ := \ 0 \ \textbf{to} \ 1 \ \textbf{do} \ y \ := \ y \ + \ 1) \ \textbf{end}) \quad (\mathbf{S}_{\text{ini}})$$

Leyenda de Estados

- $\sigma_0 \equiv \varnothing \ (\sigma_{\text{ini}})$
- $\sigma_1 \equiv \sigma_0[x \mapsto \mathcal{A}[\![1]\!]\sigma_0] = \{x \mapsto 1\}$
- $\sigma_2 \equiv \sigma_8[\text{DV}(\textbf{var} \ y \ := \ 2\,;) \longmapsto \sigma_2] = \{x \mapsto 1, \ y \mapsto 5\} \ (\sigma_{\text{fin}})$
- $\sigma_3 \equiv \sigma_2[y \mapsto \mathcal{A}[\![2]\!]\sigma_2] = \{x \mapsto 1, \ y \mapsto 2\}$
- $\sigma_4 \equiv \sigma_3[x \mapsto \mathcal{A}[\![x \ + \ 5]\!]\sigma_3] = \{x \mapsto 6, \ y \mapsto 2\}$
- $\sigma_5 \equiv \sigma_4[x \mapsto \mathcal{A}[\![0]\!]\sigma_4] = \{x \mapsto 0, \ y \mapsto 2\}$
- $\sigma_6 \equiv \sigma_5[y \mapsto \mathcal{A}[\![y \ + \ 1]\!]\sigma_5] = \{x \mapsto 0, \ y \mapsto 3\}$
- $\sigma_7 \equiv \sigma_6[x \mapsto \mathcal{A}[\![x \ + \ 1]\!]\sigma_6] = \{x \mapsto 1, \ y \mapsto 3\}$

- $\sigma_8 \equiv \sigma_7[y \mapsto \mathcal{A}[\![y\ +\ 1]\!]\sigma_7] = \{x \mapsto 1,\ y \mapsto 4\}$

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2. Semántica SOS

$$\mathcal{T}_1 = \langle x\ :=\ 1,\ \sigma_0\rangle \Rightarrow \sigma_0[x \mapsto \mathcal{A}[\![1]\!]\sigma_0]\ [\text{ass}_{\text{sos}}]$$

$$\mathcal{T}_2 = \frac{\mathcal{T}_1}{\langle (x\ :=\ 1\ ;\ y\ :=\ 5),\ \sigma_0\rangle \Rightarrow \langle y\ :=\ 5,\ \sigma_1\rangle}\ [\text{comp}^2_{\text{sos}}]$$

$$\mathcal{T}_3 = \frac{\mathcal{T}_2}{\langle S_1,\ \sigma_0\rangle \Rightarrow \langle S_2,\ \sigma_1\rangle}\ [\text{comp}^1_{\text{sos}}]$$

$$\mathcal{T}_4 = \langle y\ :=\ 5,\ \sigma_1\rangle \Rightarrow \sigma_1[y \mapsto \mathcal{A}[\![5]\!]\sigma_1]\ [\text{ass}_{\text{sos}}]$$

$$\mathcal{T}_5 = \frac{\mathcal{T}_4}{\langle S_3,\ \sigma_1\rangle \Rightarrow \langle S_4,\ \sigma_2\rangle}\ [\text{comp}^2_{\text{sos}}]$$

$$\mathcal{T}_6 = \langle \textbf{var}\ y\ :=\ 2\ ;,\ \sigma_2\rangle \Rightarrow_D \langle \varepsilon,\ \sigma_3\rangle\ \ [\text{var}_{\text{sos}}]$$

$$\mathcal{T}_7 = \langle \varepsilon,\ \sigma_3\rangle \Rightarrow_D \sigma_3\ [\text{none}_{\text{sos}}]$$

$$\mathcal{T}_8 = \frac{\mathcal{T}_6,\ \mathcal{T}_7}{\langle S_4,\ \sigma_2\rangle \Rightarrow \langle S_5,\ \sigma_3\rangle}\ [\text{begin} - \text{block}_{\text{sos}}]$$

$$\mathcal{T}_9 = \langle x\ :=\ x\ +\ 5,\ \sigma_3\rangle \Rightarrow \sigma_3[x \mapsto \mathcal{A}[\![x\ +\ 5]\!]\sigma_3]\ [\text{ass}_{\text{sos}}]$$

$$\mathcal{T}_{10} = \frac{\mathcal{T}_9}{\langle S_6,\ \sigma_3\rangle \Rightarrow \langle \textbf{for}\ x\ :=\ 0\ \textbf{to}\ 1\ \textbf{do}\ y\ :=\ y\ +\ 1,\ \sigma_4\rangle}\ [\text{comp}^2_{\text{sos}}]$$

$$\mathcal{T}_{11} = \frac{\mathcal{T}_{10}}{\langle S_5,\ \sigma_3\rangle \Rightarrow \langle S_7,\ \sigma_4\rangle}\ [\text{comp}^1_{\text{sos}}]$$

$$\mathcal{T}_{12} = \langle \textbf{for}\ x\ :=\ 0\ \textbf{to}\ 1\ \textbf{do}\ y\ :=\ y\ +\ 1,\ \sigma_4\rangle \Rightarrow \langle S_8,\ \sigma_4\rangle\ \ [\text{for}_{\text{sos}}]$$

$$\mathcal{T}_{13} = \frac{\mathcal{T}_{12}}{\langle S_7,\ \sigma_4\rangle \Rightarrow \langle S_9,\ \sigma_4\rangle}\ [\text{comp}^1_{\text{sos}}]$$

$$\mathcal{T}_{14} = \frac{\langle S_8,\ \sigma_4\rangle \Rightarrow \langle S_{10},\ \sigma_4\rangle\ \ [\text{if} - \text{tt}_{\text{sos}}]}{\langle S_9,\ \sigma_4\rangle \Rightarrow \langle S_{11},\ \sigma_4\rangle}\ [\text{comp}^1_{\text{sos}}]$$

$$\mathcal{T}_{15} = \langle x\ :=\ 0,\ \sigma_4\rangle \Rightarrow \sigma_4[x \mapsto \mathcal{A}[\![0]\!]\sigma_4]\ [\text{ass}_{\text{sos}}]$$

$$\mathcal{T}_{16} = \frac{\mathcal{T}_{15}}{\langle (x\ :=\ 0\ ;\ y\ :=\ y\ +\ 1),\ \sigma_4\rangle \Rightarrow \langle y\ :=\ y\ +\ 1,\ \sigma_5\rangle}\ [\text{comp}^2_{\text{sos}}]$$

$$\mathcal{T}_{17} = \frac{\mathcal{T}_{16}}{\langle S_{11},\ \sigma_4\rangle \Rightarrow \langle S_{12},\ \sigma_5\rangle}\ [\text{comp}^1_{\text{sos}}]$$

$$\mathcal{T}_{18} = \langle y\ :=\ y\ +\ 1,\ \sigma_5\rangle \Rightarrow \sigma_5[y \mapsto \mathcal{A}[\![y\ +\ 1]\!]\sigma_5]\ [\text{ass}_{\text{sos}}]$$

$$\mathcal{T}_{19} = \frac{\mathcal{T}_{18}}{\langle S_{13},\ \sigma_5\rangle \Rightarrow \langle \textbf{for}\ x\ :=\ x\ +\ 1\ \textbf{to}\ 1\ \textbf{do}\ y\ :=\ y\ +\ 1,\ \sigma_6\rangle}\ [\text{comp}^2_{\text{sos}}]$$

$$\mathcal{T}_{20} = \frac{\mathcal{T}_{19}}{\langle S_{12},\ \sigma_5\rangle \Rightarrow \langle S_{14},\ \sigma_6\rangle}\ [\text{comp}^1_{\text{sos}}]$$

$$\mathcal{T}_{21} = \langle \textbf{for}\ x\ :=\ x\ +\ 1\ \textbf{to}\ 1\ \textbf{do}\ y\ :=\ y\ +\ 1,\ \sigma_6\rangle \Rightarrow \langle S_{15},\ \sigma_6\rangle\ \ [\text{for}_{\text{sos}}]$$

$$\mathcal{T}_{22} = \frac{\mathcal{T}_{21}}{\langle S_{14},\ \sigma_6\rangle \Rightarrow \langle S_{16},\ \sigma_6\rangle}\ [\text{comp}^1_{\text{sos}}]$$

$$\mathcal{T}_{23} = \cfrac{\langle S_{15},\ \sigma_6\rangle \Rightarrow \langle S_{17},\ \sigma_6\rangle\ [\text{if} - \text{tt}_{\text{sos}}]}{\langle S_{16},\ \sigma_6\rangle \Rightarrow \langle S_{18},\ \sigma_6\rangle}\ [\text{comp}^1_{\text{sos}}]$$

$$\mathcal{T}_{24} = \langle x\ :=\ x\ +\ 1,\ \sigma_6\rangle \Rightarrow \sigma_6[x \mapsto \mathcal{A}[\![x\ +\ 1]\!]\sigma_6]\ [\text{ass}_{\text{sos}}]$$

$$\mathcal{T}_{25} = \cfrac{\mathcal{T}_{24}}{\langle (x\ :=\ x\ +\ 1\ ;\ y\ :=\ y\ +\ 1),\ \sigma_6\rangle \Rightarrow \langle y\ :=\ y\ +\ 1,\ \sigma_7\rangle}\ [\text{comp}^2_{\text{sos}}]$$

$$\mathcal{T}_{26} = \cfrac{\mathcal{T}_{25}}{\langle S_{18},\ \sigma_6\rangle \Rightarrow \langle S_{12},\ \sigma_7\rangle}\ [\text{comp}^1_{\text{sos}}]$$

$$\mathcal{T}_{27} = \langle y\ :=\ y\ +\ 1,\ \sigma_7\rangle \Rightarrow \sigma_7[y \mapsto \mathcal{A}[\![y\ +\ 1]\!]\sigma_7]\ [\text{ass}_{\text{sos}}]$$

$$\mathcal{T}_{28} = \cfrac{\mathcal{T}_{27}}{\langle S_{13},\ \sigma_7\rangle \Rightarrow \langle \textbf{for}\ x\ :=\ x\ +\ 1\ \textbf{to}\ 1\ \textbf{do}\ y\ :=\ y\ +\ 1,\ \sigma_8\rangle}\ [\text{comp}^2_{\text{sos}}]$$

$$\mathcal{T}_{29} = \cfrac{\mathcal{T}_{28}}{\langle S_{12},\ \sigma_7\rangle \Rightarrow \langle S_{14},\ \sigma_8\rangle}\ [\text{comp}^1_{\text{sos}}]$$

$$\mathcal{T}_{30} = \langle \textbf{for}\ x\ :=\ x\ +\ 1\ \textbf{to}\ 1\ \textbf{do}\ y\ :=\ y\ +\ 1,\ \sigma_8\rangle \Rightarrow \langle S_{15},\ \sigma_8\rangle\ [\text{for}_{\text{sos}}]$$

$$\mathcal{T}_{31} = \cfrac{\mathcal{T}_{30}}{\langle S_{14},\ \sigma_8\rangle \Rightarrow \langle S_{16},\ \sigma_8\rangle}\ [\text{comp}^1_{\text{sos}}]$$

$$\mathcal{T}_{32} = \cfrac{\langle S_{15},\ \sigma_8\rangle \Rightarrow \langle \textbf{skip},\ \sigma_9\rangle\ [\text{if} - \text{ff}_{\text{sos}}]}{\langle S_{16},\ \sigma_{10}\rangle \Rightarrow \langle S_{19},\ \sigma_8\rangle}\ [\text{comp}^1_{\text{sos}}]$$

$$\mathcal{T}_{33} = \cfrac{\langle \textbf{skip},\ \sigma_8\rangle \Rightarrow \sigma_9\ [\text{skip}_{\text{sos}}]}{\langle S_{19},\ \sigma_8\rangle \Rightarrow \langle S_{20},\ \sigma_8\rangle}\ [\text{comp}^2_{\text{sos}}]$$

$$\mathcal{T}_{34} = \langle S_{20},\ \sigma_8\rangle \Rightarrow \sigma_2\ [\text{end} - \text{block}_{\text{sos}}]$$

Sentencias Abreviadas

$$S_1 = (x\ :=\ 1\ ;\ y\ :=\ 5\ ;\ \textbf{begin var}\ y\ :=\ 2\ ;\ (x\ :=\ x\ +\ 5\ ;\ \textbf{for}\ x\ :=\ 0\ \textbf{to}\ 1\ \textbf{do}\ y\ :=\ y\ +\ 1)\ \textbf{end})\quad (\mathbf{S}_{\text{ini}})$$
$$S_2 = (\ y\ :=\ 5\ ;\ \textbf{begin var}\ y\ :=\ 2\ ;\ (x\ :=\ x\ +\ 5\ ;\ \textbf{for}\ x\ :=\ 0\ \textbf{to}\ 1\ \textbf{do}\ y\ :=\ y\ +\ 1)\ \textbf{end})$$
$$S_3 = (y\ :=\ 5\ ;\ \textbf{begin var}\ y\ :=\ 2\ ;\ (x\ :=\ x\ +\ 5\ ;\ \textbf{for}\ x\ :=\ 0\ \textbf{to}\ 1\ \textbf{do}\ y\ :=\ y\ +\ 1)\ \textbf{end})$$
$$S_4 = \textbf{begin var}\ y\ :=\ 2\ ;\ (x\ :=\ x\ +\ 5\ ;\ \textbf{for}\ x\ :=\ 0\ \textbf{to}\ 1\ \textbf{do}\ y\ :=\ y\ +\ 1)\ \textbf{end}$$
$$S_5 = (x\ :=\ x\ +\ 5\ ;\ \textbf{for}\ x\ :=\ 0\ \textbf{to}\ 1\ \textbf{do}\ y\ :=\ y\ +\ 1\ ;\ \textbf{restore}([x\ :=\ 1]\ [y\ :=\ 5],\ \textbf{var}\ y\ :=\ 2\ ;))$$
$$S_6 = (x\ :=\ x\ +\ 5\ ;\ \textbf{for}\ x\ :=\ 0\ \textbf{to}\ 1\ \textbf{do}\ y\ :=\ y\ +\ 1)$$
$$S_7 = (\textbf{for}\ x\ :=\ 0\ \textbf{to}\ 1\ \textbf{do}\ y\ :=\ y\ +\ 1\ ;\ \textbf{restore}([x\ :=\ 1]\ [y\ :=\ 5],\ \textbf{var}\ y\ :=\ 2\ ;))$$
$$S_8 = \textbf{if}\ 0\ \leq\ 1\ \textbf{then}\ x\ :=\ 0\ ;\ y\ :=\ y\ +\ 1\ ;\ \textbf{for}\ x\ :=\ x\ +\ 1\ \textbf{to}\ 1\ \textbf{do}\ y\ :=\ y\ +\ 1\ \textbf{else skip}$$
$$S_9 = (\textbf{if}\ 0\ \leq\ 1\ \textbf{then}\ x\ :=\ 0\ ;\ y\ :=\ y\ +\ 1\ ;\ \textbf{for}\ x\ :=\ x\ +\ 1\ \textbf{to}\ 1\ \textbf{do}\ y\ :=\ y\ +\ 1\ \textbf{else skip}\ ;\ \textbf{restore}([x\ :=\ 1]\ [y\ :=\ 5],\ \textbf{var}\ y\ :=\ 2\ ;))$$
$$S_{10} = (x\ :=\ 0\ ;\ y\ :=\ y\ +\ 1\ ;\ \textbf{for}\ x\ :=\ x\ +\ 1\ \textbf{to}\ 1\ \textbf{do}\ y\ :=\ y\ +\ 1)$$
$$S_{11} = (x\ :=\ 0\ ;\ y\ :=\ y\ +\ 1\ ;\ \textbf{for}\ x\ :=\ x\ +\ 1\ \textbf{to}\ 1\ \textbf{do}\ y\ :=\ y\ +\ 1\ ;\ \textbf{restore}([x\ :=\ 1]\ [y\ :=\ 5],\ \textbf{var}\ y\ :=\ 2\ ;))$$
$$S_{12} = (y\ :=\ y\ +\ 1\ ;\ \textbf{for}\ x\ :=\ x\ +\ 1\ \textbf{to}\ 1\ \textbf{do}\ y\ :=\ y\ +\ 1\ ;\ \textbf{restore}([x\ :=\ 1]\ [y\ :=\ 5],\ \textbf{var}\ y\ :=\ 2\ ;))$$
$$S_{13} = (y\ :=\ y\ +\ 1\ ;\ \textbf{for}\ x\ :=\ x\ +\ 1\ \textbf{to}\ 1\ \textbf{do}\ y\ :=\ y\ +\ 1)$$
$$S_{14} = (\textbf{for}\ x\ :=\ x\ +\ 1\ \textbf{to}\ 1\ \textbf{do}\ y\ :=\ y\ +\ 1\ ;\ \textbf{restore}([x\ :=\ 1]\ [y\ :=\ 5],\ \textbf{var}\ y\ :=\ 2\ ;))$$
$$S_{15} = \textbf{if}\ x\ +\ 1\ \leq\ 1\ \textbf{then}\ x\ :=\ x\ +\ 1\ ;\ y\ :=\ y\ +\ 1\ ;\ \textbf{for}\ x\ :=\ x\ +\ 1\ \textbf{to}\ 1\ \textbf{do}\ y\ :=\ y\ +\ 1\ \textbf{else skip}$$
$$S_{16} = (\textbf{if}\ x\ +\ 1\ \leq\ 1\ \textbf{then}\ x\ :=\ x\ +\ 1\ ;\ y\ :=\ y\ +\ 1\ ;\ \textbf{for}\ x\ :=\ x\ +\ 1\ \textbf{to}\ 1\ \textbf{do}\ y\ :=\ y\ +\ 1\ \textbf{else skip}\ ;\ \textbf{restore}([x\ :=\ 1]\ [y\ :=\ 5],\ \textbf{var}\ y\ :=\ 2\ ;))$$
$$S_{17} = (x\ :=\ x\ +\ 1\ ;\ y\ :=\ y\ +\ 1\ ;\ \textbf{for}\ x\ :=\ x\ +\ 1\ \textbf{to}\ 1\ \textbf{do}\ y\ :=\ y\ +\ 1)$$
$$S_{18} = (x\ :=\ x\ +\ 1\ ;\ y\ :=\ y\ +\ 1\ ;\ \textbf{for}\ x\ :=\ x\ +\ 1\ \textbf{to}\ 1\ \textbf{do}\ y\ :=\ y\ +\ 1\ ;\ \textbf{restore}([x\ :=\ 1]\ [y\ :=\ 5],\ \textbf{var}\ y\ :=\ 2\ ;))$$
$$S_{19} = (\textbf{skip}\ ;\ \textbf{restore}([x\ :=\ 1]\ [y\ :=\ 5],\ \textbf{var}\ y\ :=\ 2\ ;))$$
$$S_{20} = \textbf{restore}([x\ :=\ 1]\ [y\ :=\ 5],\ \textbf{var}\ y\ :=\ 2\ ;)\quad (\mathbf{S}_{\text{fin}})$$

Leyenda de Estados

- $\sigma_0 \equiv \varnothing$ (σ_{ini})
- $\sigma_1 \equiv \sigma_0[x \mapsto \mathcal{A}[\![1]\!]\sigma_0] = \{x \mapsto 1\}$
- $\sigma_2 \equiv \sigma_1[y \mapsto \mathcal{A}[\![5]\!]\sigma_1] = \{x \mapsto 1,\ y \mapsto 5\}$ (σ_{fin})
- $\sigma_3 \equiv \{x \mapsto 1,\ y \mapsto 2\}$
- $\sigma_4 \equiv \sigma_3[x \mapsto \mathcal{A}[\![x\ +\ 5]\!]\sigma_3] = \{x \mapsto 6,\ y \mapsto 2\}$
- $\sigma_5 \equiv \sigma_4[x \mapsto \mathcal{A}[\![0]\!]\sigma_4] = \{x \mapsto 0,\ y \mapsto 2\}$
- $\sigma_6 \equiv \sigma_5[y \mapsto \mathcal{A}[\![y\ +\ 1]\!]\sigma_5] = \{x \mapsto 0,\ y \mapsto 3\}$
- $\sigma_7 \equiv \sigma_6[x \mapsto \mathcal{A}[\![x\ +\ 1]\!]\sigma_6] = \{x \mapsto 1,\ y \mapsto 3\}$
- $\sigma_8 \equiv \sigma_7[y \mapsto \mathcal{A}[\![y\ +\ 1]\!]\sigma_7] = \{x \mapsto 1,\ y \mapsto 4\}$
- $\sigma_9 \equiv \{x \mapsto 1,\ y \mapsto 4\}$
- $\sigma_{10} \equiv \{x \mapsto 1,\ y \mapsto 4\}$